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Machine learning project

Title : FairCheck: A Dataset and Model-Level Bias Detection Framework

Abstract

As machine learning systems are increasingly deployed in sensitive domains like healthcare, recruitment, and finance, concerns about algorithmic bias have become critical. Models trained on historical data can inherit social prejudices, resulting in unfair treatment of individuals or groups. Detecting such bias is essential to ensure responsible AI adoption.

FairCheck is a lightweight framework designed to assess bias at both the dataset and model levels. Implemented as an interactive web application using Streamlit, it allows users to upload structured datasets and automatically evaluate fairness using metrics such as Statistical Parity Difference (SPD),

Disparate Impact (DI), and Mutual Information (MI).

Beyond dataset analysis, FairCheck supports model-level audits using classifiers like K-Nearest Neighbors (KNN) and unsupervised learning techniques. It highlights potential biases across sensitive groups through visual dashboards, summaries, and downloadable reports, bridging the gap between fairness theory and practical bias analysis.

Introduction

Machine learning (ML) has revolutionized decision-making in a wide range of domains. However, models trained on real-world data can reflect and even exacerbate existing societal biases. These hidden biases may cause disproportionate harm to certain groups particularly in high-stakes applications like criminal justice, financial lending, and hiring.

Traditional ML evaluation focuses on performance metrics like accuracy or F1-score, which do not account for disparities between groups. To address this limitation, fairness metrics have been proposed that measure how equitably a model treats different demographic segments.

FairCheck is developed as a practical and user-friendly tool that enables users to examine fairness in ML pipelines. It evaluates both datasets and models for potential bias, offering support for multiple fairness metrics, automated feature checks, and interactive visualizations all within a streamlined, low-code interface. The aim is to make fairness auditing more accessible to data scientists, students, and researchers.

Literature Review

The detection and mitigation of bias in machine learning systems has become a major focus of research in responsible AI. This section surveys ten foundational and recent research works that inform the design and motivation of FairCheck a bias detection framework at both dataset and model levels.

1. Fairness and Machine Learning: Limitations and Opportunities

Authors: Barocas, Hardt, Narayanan (2019)

This book lays the theoretical foundation for understanding fairness in ML. It highlights different fairness concepts such as group vs individual fairness and explains how they often conflict. It also emphasizes that technical solutions alone may not resolve fairness issues, pointing to deeper societal implications.

FairCheck draws from this work to shape its fairness metric selection.

2. A Survey on Bias and Fairness in Machine Learning

Authors: Mehrabi et al. (2021)

This survey reviews a wide array of literature on fairness in ML, including sources of bias, fairness definitions, and mitigation techniques. The authors classify bias into several categories and describe over a dozen fairness metrics. FairCheck uses this classification to frame its two-level bias detection pipeline.

3. Equality of Opportunity in Supervised Learning

Authors: Hardt et al. (2016)

The authors introduce fairness metrics like Equal Opportunity and Equalized Odds, which require parity in classification performance across sensitive groups. These metrics have influenced the design of FairCheck's model-level fairness checks, ensuring equal treatment in prediction outcomes.

4. Fairness Through Awareness

Authors: Dwork et al. (2012)

This paper argues that group-level fairness is not enough and introduces individual fairness — ensuring similar individuals are treated similarly. Though

difficult to implement without domain knowledge, the idea supports FairCheck's goal of interpretability and feature-level analysis.

5. Fairness-Aware Machine Learning:

A Perspective

Authors: Verma & Rubin (2018)

This paper categorizes fairness metrics and evaluates their mathematical properties. It helps distinguish between fairness in predictions and fairness in outcomes, which guides FairCheck's metric design and visualization strategy.

6. AI Fairness 360: IBM Toolkit for Fairness Auditing

Authors: Bellamy et al. (2018)

IBM's toolkit offers a wide range of bias detection and mitigation tools, both pre- and post-processing.

Although comprehensive, it has a steep learning curve. FairCheck aims to provide similar functionality in a more beginner-friendly interface.

7. Fairlearn: A Fairness-Aware Toolkit

Authors: Bird et al. (2020)

Fairlearn is designed to apply fairness constraints during model training. It integrates easily with Scikit-learn and focuses on group fairness metrics. FairCheck is inspired by Fairlearn but emphasizes visualization and modular detection first, leaving mitigation for later phases.

8. Understanding Unintended Bias in ML Systems

Authors: Suresh & Guttag (2019)

This paper outlines six types of bias in ML pipelines, from data collection to deployment. It argues that fairness must be addressed throughout the entire ML lifecycle. FairCheck adopts this holistic view by analyzing both the dataset and the trained model.

9. Discrimination in Online Ad Delivery

Author: Latanya Sweeney (2013)

This empirical study showed racial bias in Google Ads delivery based on names. It demonstrated how seemingly neutral algorithms can result in discriminatory behavior. FairCheck's emphasis on visual and measurable detection of such bias is inspired by this real-world case.

10. Fairness and Abstraction in Sociotechnical Systems

Authors: Selbst et al. (2019)

This work critiques the over-reliance on technical fixes for fairness and advocates for more contextual, human-aware solutions. It supports the need for tools like FairCheck to be transparent, interpretable, and adaptable to diverse datasets and domains.

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